Orchestrating Distributed Cloud Infrastructure with Pulumi and PyInfra.

Abstract:

In the era of multi-cloud and distributed systems, managing infrastructure often involves a fragmented mess of YAML files and specialised Domain Specific Languages (DSLs). But why compromise on logic and testability?

This session explores a unified, Python-first approach to cloud orchestration. By combining Pulumi for resource provisioning and PyInfra for high-speed, agentless configuration, we can build a robust deployment pipeline entirely in Python. We will dive into a Proof of Concept (PoC) that demonstrates how to orchestrate a distributed cluster, handle complex dependencies, and manage it all without leaving the Python ecosystem.

Key Takeaways:

- Understand the benefit of using IaPy (Infrastructure As Python Code) for the entire infrastructure lifecycle.
- Learn how to combine Cloud provisioning (Pulumi) with server configuration management (pyInfra) to solve complex deployment logic.
- Achieve High-Speed Scale: Witness how pyInfra's agentless, parallel execution engine outperforms traditional tools when managing distributed nodes.
- Implement Best Practices: Learn patterns for passing metadata between tools.

Outline: (30 mins)

- The Infrastructure "Language Soup" Problem (3 mins)
- Architectural Deep Dive (5 mins)
- Live Demo Lab: Orchestrating a Distributed Cluster (11 mins)
- Production-Ready Patterns (3 mins)
- Key takeaway and conclusion (3 mins)
- Q&A (5 mins)